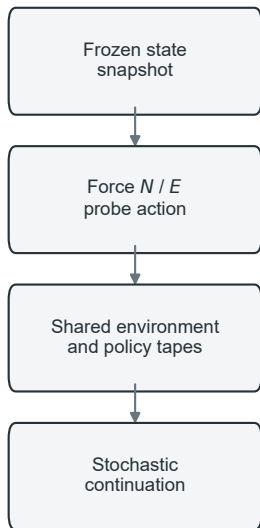
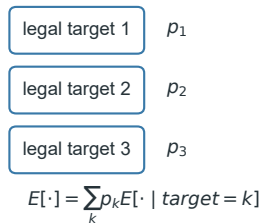


Paired counterfactual protocol and exact three-component cost reconstruction

a Snapshot + common random numbers



b Exact legal-target marginalization



c Complete cost ledger

$$\begin{aligned} &Sub_{cost, total} \\ &= Sub_{cost, same} \\ &+ Sub_{cost, future, probe} \\ &+ Sub_{cost, future, other} \end{aligned}$$

d Ledger integrity

